3. ***Multiple correlation (more than 2 variables) :*** The correlation between a group of variables and a variable which is not included in that group. It is denoted by $R_{y.(xz...)}$

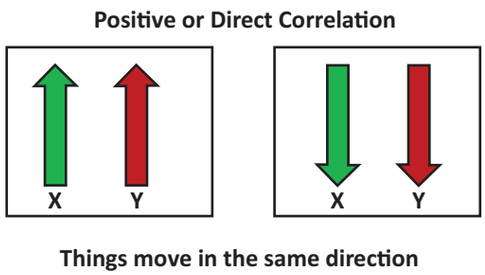
In this chapter, we study simple correlation only, multiple correlation and partial correlation involving three or more variables will be studied in higher classes .

4.2.1 Simple correlation or Linear correlation

Here, we are dealing with data involving two related variables and generally we assign a symbol 'x ' to scores of one variable and symbol 'y ' to scores of the other variable. There are five types in simple correlation. They are

1. Positive correlation (Direct correlation)
2. Negative correlation (Inverse correlation)
3. Uncorrelated
4. Perfect positive correlation
5. Perfect negative correlation

1) Positive correlation: (Direct correlation)

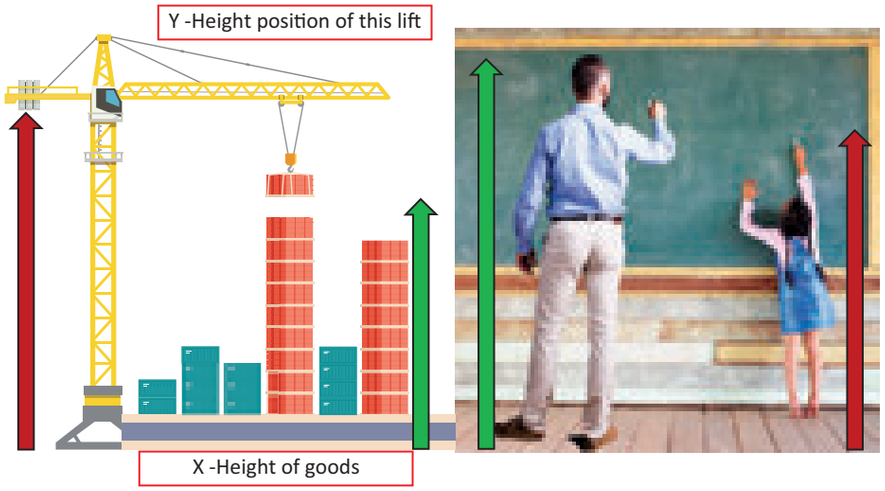


The variables are said to be positively correlated if larger values of x are associated with larger values of y and smaller values of x are associated with smaller values of y. In other words, if both the variables are varying in the *same direction* then the correlation is said to be positive. In other words, if one variable increases, the other variable (on an average) also increases or if one variable decreases, the other (on an average)variable also decreases.

For example,

i) Income and savings

ii) Marks in Mathematics and Marks in Statistics. (*i.e.*,Direct relationship pattern exists).



Height of the Lift increases / decreases according to the Height of goods increases / decreases.

The starting position of writing depends on the height of the writer.

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